IHD — Beta-Blockers & Antianginal Safety

1. Mortality shelf

WHAT YOU SEE

Top-left 'MORTALITY SHELF' with angel-winged bottles

WHAT IT ENCODES

The 'mortality shelf' holds the prognostic (life-prolonging) drugs in ischemic heart disease — those proven to improve survival, distinct from symptom-only agents. The survival quartet to memorize: Aspirin (antiplatelet), Beta-blockers (post-MI/HFrEF), ACE inhibitors/ARBs (post-MI, LV dysfunction, diabetes, HTN), and Statins. Mnemonic: 'ABCS' — Aspirin/Antiplatelet, Beta-blocker, Cholesterol (statin)/Cigarette cessation/ACEi, plus Sugar/BP control. Nitrates and ranolazine are NOT on this shelf.

BOARD TRAP

WRONG answer: prescribing a long-acting nitrate or amlodipine 'to protect the heart' for prognosis in stable angina. Discriminator: nitrates and CCBs are antianginal/symptomatic only — they do NOT reduce mortality. The drugs that prolong life are aspirin, statin, beta-blocker (esp. post-MI/HFrEF), and ACEi/ARB; the exam rewards separating prognostic from purely symptomatic therapy.

---

2. Beta-blocker: post-MI survival

WHAT YOU SEE

Angel-wing 'BETA-BLOCKER — POST-MI SURVIVAL' bottle

WHAT IT ENCODES

Beta-blockers reduce mortality after MI and in HFrEF by lowering myocardial oxygen demand — they decrease heart rate, contractility, and blood pressure, prolong diastolic coronary perfusion time, and are antiarrhythmic (raising the VF threshold). Evidence-based agents: metoprolol succinate (e.g., 25–200 mg/day), carvedilol (3.125–25 mg BID), and bisoprolol for HFrEF. In stable angina WITHOUT prior MI or HFrEF, they are mainly antianginal (symptom control), used first-line for that purpose.

BOARD TRAP

WRONG answer: stopping a beta-blocker abruptly in a stable CAD patient because they feel well. Discriminator: abrupt withdrawal causes rebound tachycardia, hypertension, and can precipitate angina, MI, or arrhythmia due to receptor up-regulation — taper over ~1–2 weeks. Also wrong is withholding beta-blockade post-MI without a true contraindication, since it is a proven survival therapy.

---

3. Aspirin: ↓coronary events

WHAT YOU SEE

'ASA EMPIRE — CORONARY EVENTS' bottle

WHAT IT ENCODES

Aspirin irreversibly inhibits COX-1, blocking thromboxane A2-mediated platelet aggregation, and reduces coronary events and mortality in established CAD/post-MI — a prognostic therapy. Secondary-prevention dose is 75–100 mg/day (81 mg in the US) indefinitely. After ACS or PCI, dual antiplatelet therapy (aspirin + a P2Y12 inhibitor such as clopidogrel, ticagrelor, or prasugrel) is used for a defined period.

BOARD TRAP

WRONG answer: routinely starting aspirin for PRIMARY prevention in a low-risk patient without established CAD. Discriminator: in established CAD aspirin clearly reduces events (give it), but for primary prevention the bleeding risk often offsets benefit and guidelines no longer recommend it broadly — the survival benefit is established in SECONDARY, not routine primary, prevention.

---

4. Statin: mortality regardless of LDL

WHAT YOU SEE

'STATIN — MORTALITY REGARDLESS OF LDL' bottle

WHAT IT ENCODES

High-intensity statins (atorvastatin 40–80 mg, rosuvastatin 20–40 mg) reduce mortality and recurrent events in CAD regardless of baseline LDL, via LDL lowering plus pleiotropic effects (plaque stabilization, anti-inflammatory, endothelial). All patients with clinical ASCVD get a high-intensity statin; add ezetimibe and/or a PCSK9 inhibitor if LDL stays above goal (typically <55–70 mg/dL in very-high-risk ASCVD).

BOARD TRAP

WRONG answer: withholding or stopping a statin in a CAD patient because the LDL is already 'normal.' Discriminator: statin benefit in established CAD is independent of starting LDL — every patient with clinical ASCVD warrants a high-intensity statin for its mortality/plaque-stabilizing effect; a normal LDL is not a reason to omit it.

---

5. Nitrate: symptoms only

WHAT YOU SEE

'NITRATE — SYMPTOMS ONLY, NO SURVIVAL DATA' bottle

WHAT IT ENCODES

Nitrates (sublingual nitroglycerin for acute angina; isosorbide mono/dinitrate for prophylaxis) relieve angina by venodilation (reducing preload and wall stress, lowering O2 demand) and some coronary vasodilation — but have NO proven mortality benefit in stable IHD. Provide a nitrate-free interval (~10–14 h/day) to prevent tachyphylaxis/tolerance.

BOARD TRAP

WRONG answer: co-prescribing a nitrate with a PDE5 inhibitor (sildenafil, tadalafil, vardenafil). Discriminator: the combination causes profound, potentially fatal hypotension via additive cGMP-mediated vasodilation — nitrates are contraindicated within 24 h of sildenafil/vardenafil and 48 h of tadalafil. Also remember nitrates do NOT prolong life; they are symptomatic only.

---

6. BB + non-DHP CCB: never together

WHAT YOU SEE

Knight vs soldier fighting, 'NEVER TOGETHER', 'USE WITH CAUTION'

WHAT IT ENCODES

Beta-blockers plus non-dihydropyridine CCBs (verapamil, diltiazem) cause ADDITIVE suppression of the SA and AV nodes — both slow the heart rate and AV conduction and depress contractility — risking severe bradycardia, high-grade AV block, and decompensated heart failure. Avoid this combination, especially in patients with conduction disease or LV dysfunction.

BOARD TRAP

WRONG answer: adding verapamil or diltiazem to a beta-blocker for better rate/angina control. Discriminator: this stacks negative chronotropy/dromotropy and inotropy → heart block and bradycardia. The SAFE add-on antianginal alongside a beta-blocker is a DIHYDROPYRIDINE CCB (amlodipine), which has minimal AV-nodal effect.

---

7. BB avoid in asthma/COPD

WHAT YOU SEE

'ASTHMA/COPD = AVOID', propranolol with red X over lungs

WHAT IT ENCODES

Non-selective beta-blockers (propranolol, nadolol, carvedilol) block bronchial β2 receptors and can precipitate bronchospasm — avoid or use with caution in asthma and severe/reactive COPD. They also blunt β2-mediated recovery from hypoglycemia and mask adrenergic warning signs (tachycardia, tremor), a hazard in insulin-treated diabetics.

BOARD TRAP

WRONG answer: choosing propranolol (or any non-selective beta-blocker) when a beta-blocker is needed in a patient with asthma. Discriminator: non-selective agents trigger bronchospasm; if a beta-blocker is truly indicated (e.g., post-MI), use a CARDIOSELECTIVE β1 agent (metoprolol, bisoprolol, atenolol) at the lowest effective dose with monitoring.

---

8. B1-selective = lung safe

WHAT YOU SEE

'B1 ONLY — LUNG SAFE', green-check bottles + lungs

WHAT IT ENCODES

Cardioselective (β1-selective) beta-blockers — metoprolol, bisoprolol, atenolol, nebivolol — preferentially block cardiac β1 receptors with relatively little β2 (bronchial) effect, so they are comparatively lung-safe when a beta-blocker is needed in mild-to-moderate airway disease. Note: selectivity is dose-dependent and is LOST at high doses, so β2 blockade can re-emerge.

BOARD TRAP

WRONG answer: assuming a 'cardioselective' beta-blocker is completely safe at any dose in airway disease. Discriminator: β1 selectivity is relative and dose-dependent — at high doses metoprolol/atenolol also block β2 and can provoke bronchospasm. Use the lowest effective dose, monitor, and still avoid in severe/brittle asthma.

---

9. Amlodipine: no AV effect

WHAT YOU SEE

Blue 'AMLODIPINE' hero, 'DHP = NO AV EFFECT'

WHAT IT ENCODES

Amlodipine is a dihydropyridine (DHP) CCB that acts mainly on vascular smooth muscle (arterial vasodilation, reducing afterload and relieving angina) with negligible effect on the SA/AV node — so it does NOT cause meaningful bradycardia or AV block. This makes it safe to combine with a beta-blocker, which offsets any reflex tachycardia from the vasodilation.

BOARD TRAP

WRONG answer: avoiding amlodipine alongside a beta-blocker out of fear of heart block. Discriminator: unlike verapamil/diltiazem (non-DHP, AV-nodal blockers), amlodipine has no clinically important AV-nodal effect — the beta-blocker + DHP combination is a recognized SAFE antianginal pairing. (DHPs can cause peripheral edema and reflex tachycardia when used alone.)

---

10. Carvedilol + amlodipine = safe combo

WHAT YOU SEE

'CARVEDILOL' hero, 'SAFE COMBO' handshake

WHAT IT ENCODES

Combining a beta-blocker (e.g., carvedilol — a non-selective β-blocker with added α1-blockade and proven HFrEF survival benefit) with a DHP CCB (amlodipine) is a safe, effective antianginal strategy: the beta-blocker controls heart rate/contractility and demand while amlodipine lowers afterload and provides coronary vasodilation, and the beta-blocker prevents amlodipine's reflex tachycardia.

BOARD TRAP

WRONG answer: pairing carvedilol with verapamil/diltiazem for the same effect. Discriminator: the SAFE combination is beta-blocker + DHP (amlodipine); beta-blocker + non-DHP (verapamil/diltiazem) stacks AV-nodal/contractility suppression and risks bradycardia, heart block, and HF — pick the DHP when adding a CCB to a beta-blocker.

---

11. Use with caution: normal LV

WHAT YOU SEE

Yellow caution tape 'USE WITH CAUTION — NORMAL LV, NO CONGESTION'

WHAT IT ENCODES

Even in patients with normal LV function and no congestion, rate-lowering agents (beta-blockers, non-DHP CCBs, combinations, IV agents) require caution — titrate to avoid symptomatic bradycardia, hypotension, and conduction disturbance. Check the baseline ECG (PR interval, AV block) and resting heart rate before and during up-titration; the absence of HF does not eliminate conduction risk.

BOARD TRAP

WRONG answer: aggressively up-titrating rate-control drugs in a patient with a borderline-low heart rate or first-degree AV block just because LV function is normal. Discriminator: bradycardia and AV block depend on conduction-system status, not only on LV/congestion — a normal ejection fraction does not protect against drug-induced heart block; assess the ECG and heart rate first.

---

12. IV beta-blocker in HR 104 patient

WHAT YOU SEE

Bedside monitor 'HR 104', 'IV BETA-BLOCKER' drip

WHAT IT ENCODES

IV beta-blockers (e.g., metoprolol IV, or short-acting esmolol) can be used for acute rate and ischemia control in ACS/tachycardia to cut myocardial O2 demand — but only when hemodynamically appropriate. Contraindications: acute decompensated/congested heart failure, hypotension or cardiogenic shock, significant bradycardia or high-grade AV block, and active bronchospasm.

BOARD TRAP

WRONG answer: giving an IV (or any) beta-blocker for COCAINE-induced MI/chest pain, or to a hypotensive/decompensated-HF patient. Discriminator: in cocaine toxicity, beta-blockade leaves alpha-adrenergic stimulation unopposed → worsened coronary vasoconstriction and hypertension ('unopposed alpha') — use benzodiazepines, nitrates, and CCBs instead. Likewise, never give a beta-blocker in cardiogenic shock, decompensated HF, hypotension, or bradycardia/AV block.